

# Dig Dug Repair Lab — Task Code

## Task 1: Fix BFS pathfinding

In `bfs_path()`, remove the oldest queued cell first. Replace `queue.pop()` with `queue.popleft()`. Keep the rest of your existing function unchanged.

```
current = queue.popleft()
```

BFS requires FIFO queue behavior. Using `pop()` at the right end can produce stack-like, DFS-style exploration.

## Task 2: Implement `dirt_color(row)`

```
def dirt_color(row):
    if row < 3:
        return (194, 157, 105) # Light sandy soil
    elif row < 6:
        return (160, 110, 65)  # Brown soil
    elif row < 9:
        return (120, 75, 45)   # Dark brown soil
    else:
        return (85, 50, 35)    # Deep underground soil
```

The thresholds assume the game has enough rows to show all four bands. Adjust them if your grid has a different height.

## Task 3: Implement `on_enemy_popped(enemy, score)`

```
def on_enemy_popped(enemy, score):
    row = enemy.cell[0]
    bonus = row * 50

    print(f"Enemy popped at depth {row}!")
    print(f"Bonus points: {bonus}")      print(f"Current
score: {score + bonus}")
```

## Task 4: Implement `enemy_speed_multiplier(level)`

```
def enemy_speed_multiplier(level):
    return 1 + 0.1 * level
```

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